

EGR 3350-07

Technical Communications for Engineers and Computer Scientists

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Individual Portfolio

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Computer Science

April 30, 2026

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1. Mechanism Description

Revision Plan

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Revision Plan: Wireless Fast Phone Charger

Overview of Original Feedback

The Mechanism Description received a score of 6.5 out of 7.0 (A). The instructor identified several recurring issues: (1) overuse of contractions, which are inappropriate for technical writing; (2) imprecise pronoun use, particularly sentences beginning with 'it' or 'it's' where the subject should be explicitly named; (3) informal word choices and constructions that do not suit technical description; and (4) sentences flagged as 'unclear' or attempting to do 'too much at once.' The revision addresses each of these issues systematically across all sections.

Planned Changes by Section

Introduction

The introduction will replace all contractions with full forms (e.g., 'It's' becoming 'It is') and refine the measurement of phrasing to read 'measuring approximately 10 cm in diameter and 0.8 cm in height. The pronoun 'its,' when used to open a clause about the charger's surface, will be clarified so the grammatical subject is the charger itself.

Charging Surface

This section received the most feedback. The instructor flagged the use of 'It's' (a contraction), the informal phrase 'so that the,' and noted that contractions are 'ok, but not for the subject of a sentence.' The revision rewrites every sentence using 'The charging surface is...' as the consistent grammatical subject, eliminates all contractions, and splits any compound sentence flagged as doing 'too much at once' into shorter, direct statements.

Indicator Light Ring

The phrase beginning with 'So it glows' was flagged as unclear and informal. The revision rewrites this to explicitly state that the indicator light ring illuminates in different colors

to communicate charging status, removing the informal connector and making the signaling mechanism precise.

Protective Casing

The instructor struck through the informal description 'It's a tough plastic enclosure' and indicated cleaner phrasing. The revision adopts the instructor's suggestion: 'It is made of tough plastic and is approximately 9.5 cm wide and 0.8 cm high.' All remaining informal phrasing in this section is replaced.

Power Input Port

The instructor recommended 'It is approximately' to replace the original phrasing, and flagged 'so that the user' as informal. The revision adopts the instructor's phrasing: 'The port is approximately 8 mm wide,' and replaces 'so that' constructions with precise technical alternatives such as 'allowing the user to...' or 'enabling repeated daily use without...'

Conclusion

The instructor advised to 'use short, direct sentences' and avoid trying to do too much at once. The conclusion is rewritten as a sequence of clear, direct statements — each sentence naming one component and its function without embedding multiple ideas in a single clause.

General Style Changes Across All Sections

The revision makes the following changes throughout: (1) eliminates all contractions, replacing them with full forms; (2) begins each component description with the full component name as the grammatical subject rather than a pronoun; (3) breaks any sentence flagged as 'unclear' or 'too much at once' into two shorter sentences; and (4) replaces informal connectors such as 'so' and 'so that' with precise technical language.

Revised Mechanism Description

EGR 3350-07

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Wireless Fast Phone Charger

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Computer Science

April 30, 2026

Introduction

A *wireless fast phone charger* is a power-transfer device that supplies electrical energy to a smartphone without a direct, wired connection. Energy transfers through electromagnetic induction when a compatible phone is placed on the charger's upper surface. The charger is circular, black, and constructed of rigid polymer plastic, measuring approximately 10 cm in diameter and 0.8 cm in height. The mechanism consists of four observable components: the charging surface, the indicator light ring, the protective casing, and the power input port.

Discussion

Figure 1 shows the wireless fast phone charger and its primary components.

Charging Surface

The *charging surface* is a circular support platform that stabilizes a smartphone during energy transfer. It is a flat, round platform approximately 10 cm across and 0.8 cm thick. The surface has a matte finish that provides friction, preventing the phone from sliding during charging. The flat shape positions the device consistently over the induction coil beneath, ensuring reliable energy transfer. Around the outer edge of the charging surface runs the indicator light ring.

Indicator Light Ring

The *indicator light ring* is a visual signaling component that communicates charging status to the user. It has a diameter of approximately 0.2 cm and is constructed of translucent plastic. The indicator light ring illuminates in different colors depending on whether the device is actively charging. When a phone is charging, the ring emits a steady light; when no device is detected, the ring remains unlit. The indicator light ring encircles the charging surface and rests on top of the protective casing.

Protective Casing

The *protective casing* is a structural enclosure that houses and supports the internal electrical components, including the induction coil and supporting electronics. It is made of tough plastic and measures approximately 9.5 cm wide and 0.8 cm high. The casing shields the internal components from physical impact and environmental exposure. Small rubber pads on the bottom surface of the casing prevent the charger from sliding on flat surfaces. The power input port is integrated into the side of the protective casing.

Power Input Port

The *power input port* is a connector that enables the charger to receive power from an external source. The port is located on the side of the casing and is approximately 8 mm wide, accommodating a standard USB-C plug. The enclosure surrounding the port is constructed of durable material, allowing repeated daily connection and disconnection without degradation. Once connected, power flows directly into the induction system housed within the casing.

Conclusion

A *wireless fast phone charger* is a power-transfer device that supplies electrical energy to a smartphone without a direct, wired connection. The charging surface holds the device in the correct position over the induction coil. The indicator light ring signals the current charging status to the user. The protective casing encloses and shields the internal electronics. The power input port connects the unit to an external power source. Together, these four components enable

fast, reliable wireless charging.

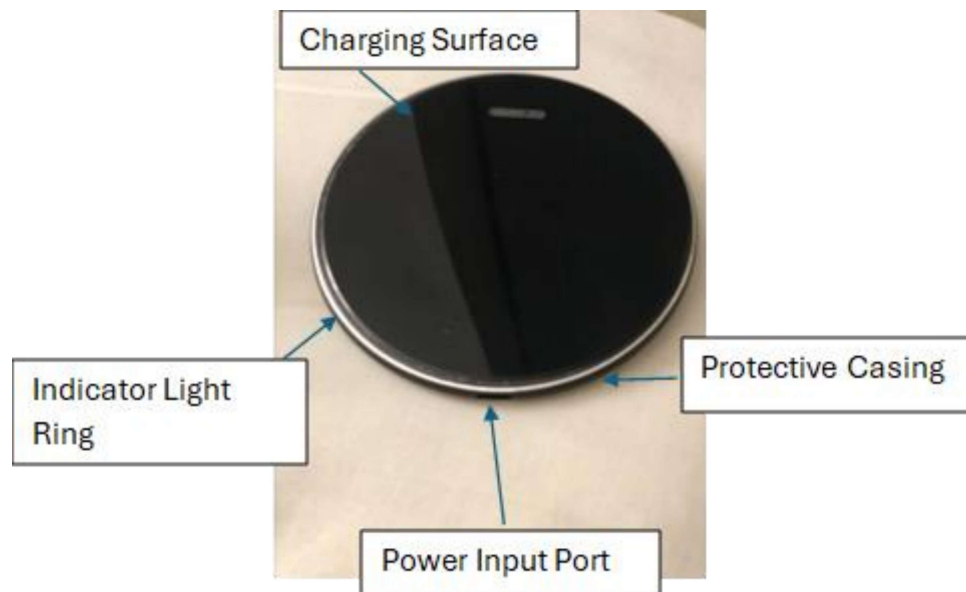


Figure 1. Wireless Fast Phone Charger, showing labeled components. Source: Author.